

NETWORKING HANDS-ON – DAY 1

Cisco Packet Tracer Basic Network Configuration & Device Exploration

1. AIM

To install Cisco Packet Tracer and design basic network topologies including direct connection, switch-based network, star topology, and to explore networking devices such as PC, Switch, Router, and Server. Also to verify connectivity using the ping command.

2. REQUIREMENTS

- Cisco Packet Tracer Software
 - Windows Laptop / PC
 - Basic understanding of IP addressing
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3. THEORY

A computer network is a collection of interconnected devices that communicate and share data. The main devices used in networking are:

- **PC (Personal Computer):** End device used by users to send and receive data
 - **Switch:** Connects multiple devices within the same local network (LAN)
 - **Router:** Connects different networks and acts as a gateway
 - **Server:** Provides services such as data, websites, and applications
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4. EXPERIMENTS AND PROCEDURE

EXPERIMENT 1: Direct PC to PC Connection

Procedure:

1. Open Cisco Packet Tracer
2. Add 2 PCs into workspace
3. Connect both PCs using **Copper Cross-over cable**
4. Assign IP addresses manually

IP Configuration:

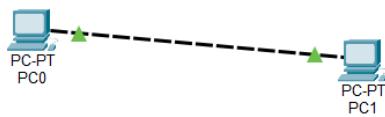
- PC0: 192.168.1.1 / 255.255.255.0
- PC1: 192.168.1.2 / 255.255.255.0

Testing:

ping 192.168.1.2

Observation:

Both PCs communicated successfully when configured in the same network.



EXPERIMENT 2: Switch-Based Network**Procedure:**

1. Add 2 PCs and 1 Switch (2960)
2. Connect PCs to switch using **Copper Straight-through cable**
3. Assign IP addresses

IP Configuration:

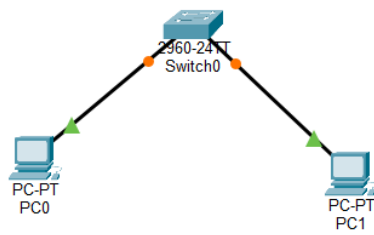
- PC0: 192.168.1.1
- PC1: 192.168.1.2

Testing:

ping 192.168.1.2

Observation:

Switch successfully enabled communication between multiple devices in the LAN.



EXPERIMENT 3: Star Topology

Procedure:

1. Add 1 Switch and 4 PCs
2. Connect all PCs to switch (Star topology)
3. Assign IP addresses to all PCs

IP Configuration:

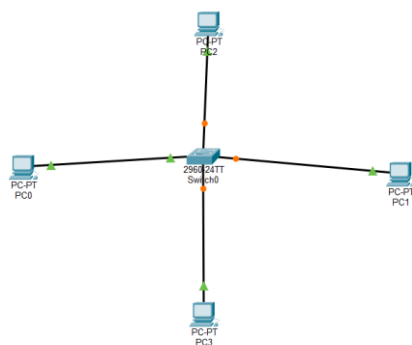
- PC1: 192.168.1.1
- PC2: 192.168.1.2
- PC3: 192.168.1.3
- PC4: 192.168.1.4

Testing:

Ping between multiple PCs.

Observation:

All devices communicated through a central switch, forming a star topology.



EXPERIMENT 4: Device Exploration and Router Configuration

Devices Used:

- PC
 - Switch
 - Router
 - Server
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Router Configuration (CLI):

```
enable
configure terminal
interface gigabitEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

PC Configuration:

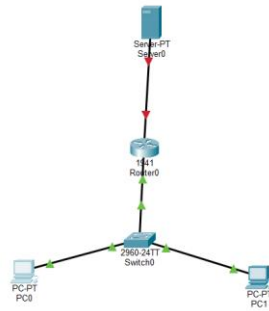
- IP Address assigned manually
 - Subnet Mask configured
 - Default Gateway: 192.168.1.1
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Testing:

```
ping 192.168.1.1
```

Observation:

Router successfully communicated with PCs after proper configuration and activation of interface.



5. OBSERVATIONS

- Devices must be in the same network to communicate directly
- Switch enables communication within LAN
- Router connects different networks and acts as a gateway
- IP address and subnet mask are essential for communication
- ping command is used to test connectivity

6. CONCLUSION

The experiment successfully demonstrated basic networking concepts using Cisco Packet Tracer. Different network topologies were created, devices were configured with IP addresses, and connectivity was verified using the ping command. This helped in understanding the role of PC, Switch, Router, and Server in a network.